

SENSOR ARCHITECTURE — BROAD PHYSIOLOGICAL ACQUISITION LAYER

GENERAL SENSOR FRAMEWORK

Node 5 may incorporate one or more physiological sensing subsystems configured to acquire systemic biometric data representing contextual physiological state. The disclosed sensor architecture is intentionally modality-agnostic and encompasses any sensing technology capable of measuring bodily signals relevant to interpretation of intimate physiology.

CARDIOVASCULAR SIGNAL SENSING

Sensors may include photoplethysmography devices, electrocardiography electrodes, ballistocardiography sensors, pulse wave analysis components, bioimpedance cardiography circuits, acoustic cardiac sensors, or equivalent technologies configured to detect cardiovascular activity, vascular dynamics, and circulatory readiness indicators.

AUTONOMIC AND NEUROPHYSIOLOGICAL SIGNALS

Node 5 may acquire signals associated with autonomic nervous system activity including electrodermal activity sensors, galvanic skin response detectors, neural interface sensors, peripheral nerve activity detectors, thermoregulation monitoring components, or other technologies capable of detecting sympathetic and parasympathetic activation patterns.

RESPIRATORY AND MOTION SENSING

Respiration-related measurements may be obtained using bioimpedance respiration sensing, strain gauges, pressure sensors, accelerometers, gyroscopes, inertial measurement units, acoustic breathing sensors, or motion-derived respiratory inference models. Motion sensing may further enable posture detection, activity classification, and artifact rejection.

THERMAL AND ENVIRONMENTAL SENSING

Temperature sensors, heat flux sensors, infrared detectors, humidity sensors, barometric pressure sensors, ambient light sensors, and environmental monitoring components may be incorporated to provide contextual information describing environmental influences on physiological state.

MULTI-MODAL SENSOR FUSION

Signals from multiple sensing modalities may be combined through sensor fusion algorithms configured to generate composite contextual physiology metrics. Fusion may occur on-device, at an edge processor, or within cloud-based computational infrastructure and may utilize statistical modeling, adaptive filtering, machine learning, or artificial intelligence techniques.

FUTURE SENSOR TECHNOLOGY COVERAGE

The disclosed sensor architecture explicitly includes sensing technologies developed after the filing date of the present disclosure, including quantum sensors, biochemical sensing platforms, optical imaging modalities, implant-adjacent sensing systems, or other emerging physiological acquisition methods capable of performing substantially equivalent contextual monitoring functions.